MIL-B-3067B
22 JUNE 1959

SUPERSEDING
MIL-R-3067A
19 APRIL 1957

MILITARY SPECIFICATION

BRIDGE, RESISTANCE (RESISTANCE BRIDGE, ZM-4( )/U)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of portable, battery-operated, dial-decade Wheatstone Bridge designated as Resistance Bridge ZM-4( )/U which may be used for locating faults in communication circuits or for making resistance measurements (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-P-515 — Plywood, Container Grade.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior, Packaging.

MIL-T-945 — Test Equipment, for Use with Electronic Equipment; General Specification.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-DL-99637 — Resistance Bridge ZM-4( )/U.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

FSC 6625

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2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. POST OFFICE

Parcel Post Regulations.

(Application for copies should be addressed to any first class Post Office.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification 4.

(Application for copies of the above publication should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification No. A–3 and No. 13.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1424 Sixteenth Street, N.W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 **Construction.** Resistance Bridge ZM–4( )/U shall be constructed in accordance with Drawing and Data List SC–DL–99637.

3.2 **Parts, materials, and processes; general.** In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL–T–945 including the selection requirements therein, as applicable. In case of conflict between the requirements of this specification and Specification MIL–T–945, this specification shall govern.

3.3 **Detail requirements.**

3.3.1 *Rheostat coils.* Manganin wire, Karma wire, or similar wire, having a negligible temperature coefficient of resistance shall be used in the rheostat and the multiplier. The coils shall be so wound as to minimize inductive or capacitive effects. All coils shall be thoroughly impregnated with a moisture-repelling compound. They shall be so treated as to retain their resistance values permanently. The resistors shall be wound on steatite forms.

3.3.1.1 *Accuracy of coils.* When tested as specified in 4.8.1, the resistances of the rheostat coils shall be within 0.1 percent of their rated resistances, and resistances of the multiplier coils within 0.05 percent, at 23 ± 3 C.

3.3.1.2 *Effect of 0.5 watt loading.* When tested as specified in 4.10, each coil shall be capable of dissipating the power of 0.5 watt without damage to itself or other parts of the set.

3.3.2 *Rheostat and multiplier switches.* The terminal studs shall be of solid nickel or nickel-plated brass; the brushes shall be laminated and made of high-grade phosphor bronze, and shall be so shaped as to make a positive low-resistance contact with the studs and the center bushing. The switches shall turn in either direction from any setting, and the proper contact between a brush and any stud shall be indicated by an audible click or by other equally effective means. Switch movement shall have positive detent action.

3.3.2.1 *Accuracy of rheostat and multiplier switches.* When tested as specified in 4.8.2, the readings shall differ from the corresponding setting of the standard dial resistance box by not more than 0.15 percent ± 0.01 ohm at the dial setting plus the resistance of test leads.

3.3.3 *Galvanometer.* The test set shall be equipped with a D'Arsonval-type galvanometer. The galvanometer shall be of the suspended-coil type and its sensitivity shall be such that 1 microampere or less gives 1 millimeter scale deflection (see 3.3.3.1 and 4.9.1). The galvanometer shall be a replaceable unit; it shall have a device for clamping both coil and needle, and means shall be provided for adjusting the zero setting. The galvanometer needle shall be protected by a glass cover of sufficient size to allow a full view of the scale which shall be suitably divided and clearly marked.

3.3.3.1 *Sensitivity.* When tested as specified in 4.9.1, the deflection of the galvanometer for the most sensitive setting of the shunt shall be not less than 3.0 millimeters of the scale.

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3.3.3.2 *Response.* When tested as specified in 4.9.2, the galvanometer period shall not exceed 3.0 seconds.

3.3.3.3 *Dead beat.* When tested as specified in 4.9.3, the galvanometer shall be dead beat (no movement of the needle).

3.3.4 *Murray and Varley loop operation.* The equipment shall be capable of operation as a Murray and Varley loop tester.

3.4 **Service conditions.** The service condition requirements shall be in accordance with the service condition requirements of Specification MIL-T-945.

3.4.1 *Vibration, bounce, and shock.*

3.4.1.1 *Vibration.* The bridge shall be tested in accordance with Specification MIL-T-945 and shall suffer no damage or failure in specification performance.

3.4.1.2 *Bounce.* Full specification performance. No physical damage except surface abrasion (see 4.13).

3.4.1.3 *Shock; ballistic.* Components shall be operable. Minor mechanical damage is permissible (see 4.12).

3.4.1.4 *Shock; bench-handling.* Full specification performance. No physical damage (see 4.12.1).

3.4.2 *Immersion.* There shall be no evidence of leakage when tested in accordance with 4.12.2.

3.5 **Preproduction samples.** The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract (see 6.2(e) (1)).

3.6 **Preconditioning.** Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.5.

3.7 **Technical literature, tools, and running spare parts.** Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order (see 6.2).

3.8 **Interchangeability.** Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used (see 4.14). When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.9 **Workmanship.** The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-T-945:
General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Flux and cleaning agents for soldering.

(b) In this specification:
3.1 Construction.
3.3.1 Rheostat coils.
3.3.2 Rheostat and multiplier switches.
3.3.3 Galvanometer.
4.1 Visual-mechanical.
Process for soldering.
Cleaning of equipment.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Mechanical operation of controls.
Gears.
Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery (see 4.3).

(b) Acceptance inspection. (Acceptance inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.15).

(1) Acceptance inspection of items before preparation for delivery (see 4.4).

(2) Acceptance inspection of preparation for delivery (see 4.15 and section 5).

4.3 Preproduction inspection. Preproduction inspection shall consist of the group A inspection and nondestructive group B and group C inspection specified in tables II, III, and IV, respectively, and the preproduction inspection specified in table I. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) shock; bench-handling, (3) shock; ballistic, and (4) immersion. Other preproduction inspection may precede, follow, or be interspersed between the foregoing. Failure of a bridge in any inspection or test shall result in withholding of preproduction approval.

TABLE I. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.3)	Requirement paragraph	Inspection paragraph
Altitude	3.4	4.12
Humidity	3.4	4.12
Fungi	3.4	MIL-T-945
Temperature	3.4	4.12
Vibration	3.4	4.12
Shock, bench-handling	3.4	4.12.1
Salt spray	3.4	4.12
Shock, ballistic	3.4	4.12
Bounce	3.4	4.13

4.4 Acceptance inspection of items before preparation for delivery. The minimum inspection performed by the contractor shall be as specified by (a) through (d) below — to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government—at its discretion— will verify, by product inspection, the contractor's compliance with specified requirements (see 6.5). Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Acceptance inspection shall consist of group A, group B, and group C inspection as specified in 4.4.1 through 4.4.3.2.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Govment.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied

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when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly (see 3.6).

4.4.1 *Group A inspection.* This inspection (including sampling) shall conform to table II and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Components				
Visual and mechanical				
Rheostat coils	3.3.1	4.6	1%	4%
Multiplier coils	3.3.1	4.6	1%	4%
Electrical				
Accuracy of rheostat coils	3.3.1.1	4.8.1	1%	(1)
Accuracy of multiplier coils	3.3.1.1	4.8.1	1%	(1)
Resistance Bridge ZM–4( )/U				
Visual and mechanical				
Bridge	3.9	4.6	1.5 dphu	6.5 dphu
Case	3.9	4.6	1.0%	4%
Electrical				
Accuracy of bridge	3.3.2.1	4.8.2		(1)
Galvanometer			2.5 dphu	
Sensitivity	3.3.3.1	4.9.1	for group	(1)
Response	3.3.3.2	4.9.2		(1)
Dead beat	3.3.3.3	4.9.3		(1)
Operational test				
Murray and Varley loop operation	3.3.4	4.11	1%	(1)

[1] All electrical defects are considered major.

4.4.2 *Group B inspection.* This inspection (including sampling) shall conform to table III and the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.4.5 and the requirements of Standard MIL–STD–105 for disposition of rejected product.

TABLE III. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL
Immersion	3.4.2	4.12.2	6.5%

4.4.2.1 *Group B sampling plan.* The group B sampling plan for the AQL listed in table III, shall be as follows:

Group B plan	*AQL*	*Inspection level for normal inspection*	*Inspection level for reduced inspection*
B–2	6.5%	L–7	L–5

4.4.2.2 *Procedure in case of failure.* When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall

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be referred to the contracting officer for resolution.

4.4.3 *Group C inspection.* Group C inspection shall be as listed in table IV.

4.4.3.1 *Sampling for inspection of equipment.* Five units per 300 or fraction thereof produced shall be selected without regard to their quality for group C inspection.

4.4.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrosive action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE IV. *Group C inspection*

Inspection	Requirement paragraph	Inspection paragraph
0.5 watt loading	3.3.1.2	4.10

4.4.4 *Reinspection of conforming group B and group C sample units.* Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.4.5 *Disposition of nonconforming product.* The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 **Bounce preconditioning.** The equipment with the galvanometer needle and coil clamped by its clamping device and with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the package tester as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.6 **Visual and mechanical inspection.** The equipment shall be examined for the defects listed in Standard MIL-STD-252 and table V.

4.7 **Inspection conditions.** Unless otherwise specified herein, all measurements and tests shall be performed at a temperature of 25° C. +10° C. —5° C. (77° F. +18° F. —9° F), at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity no greater than 80 percent.

TABLE V. *Classification of visual and mechanical defects*

Classification	
Major	Missing or broken stay chain.
	Carrying handles, hinges, or lid fasteners attached to wrong side of case.
	Missing or inoperative cover.
Minor	Cover reinforcements missing or improperly assembled.
Control	Difficulty in operation of carrying handles or cover.

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4.8 Accuracy. The following tests shall be made for accuracy using a dial resistance box of known resistance standards certified by the Bureau of Standards for an accuracy of at least 10 times better than that specified for the coils specified in 3.3.1.1. All tests shall be made on the completed bridge except for the accuracy of coils test specified in 4.8.1, which shall be made on the individual coils prior to assembly into the bridge.

4.8.1 *Accuracy of coils.* The coils of the bridge shall be connected across a power source so as to dissipate a maximum of 0.3 watts of power. The coils shall then be measured at ambient temperature against the known resistance standards, and shall meet the requirements of 3.3.1.1 for accuracy.

4.8.2 *Accuracy of rheostat and multiplier switches.* Rheostat and multiplier (ratio arm) switch shall be tested by measuring the known resistance standards with the bridge. The rheostat switches shall be measured at 10 settings throughout the range. The requirements of 3.3.2.1 shall be met.

4.9 Galvanometer. The galvanometer assembled in the bridge shall be tested as follows:

4.9.1 *Sensitivity test.* The lever key to adapt the bridge for the Murray loop test shall be thrown and rheostat shall be set to infinity. Four and one-half (4.5) volts of direct current shall then be applied in series with a one-megohm resistor across binding posts (usually marked "X") to which the unknown resistance is normally connected. The requirements of 3.3.3.1 shall be met.

4.9.2 *Response test.* The galvanometer period shall be tested by timing the damping of the needle when reading a known resistance standard, and shall meet the requirements of 3.3.3.2.

4.9.3 *Deadbeat test.* The clamping action of the device for holding the needle and coil shall be tested and shall meet the requirements of 3.3.3.3.

4.10 Effect of 0.5 watt loading test. The resistance coils shall be loaded for 6 hours to dissipate at least 0.5 watts of power, and shall meet the requirements of 3.3.1.2.

4.11 Murray and Varley loop operation. The equipment shall be tested for Murray and Varley loop operation in accordance with Drawing SC–D–99679, a detail of Drawing SC–DL–99637.

4.12 Service condition tests. The preproduction samples shall be tested in accordance with service conditions test of Specification MIL–T–945, for compliance with the requirements of 3.4 of this specification.

4.12.1 *Shock test; bench-handling.* The chassis-and-front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30°, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.12.2 *Immersion test.* The test set, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40° F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within 2° F, of its initial

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temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. The equipment shall meet the requirements of 3.4.2.

4.13 **Bounce test.** The equipment, in its transit case (if any), shall be placed on the table of the package tester as made by the L.A.B. Corporation, Skaneateles, N. Y., or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm $\pm$ 1 percent, for a total of 3 hours. During each ½-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.14 **Inspection for interchangeability.** The dimensions listed below shall be gaged or assured to determine compliance with the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts of assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(e) Location of connectors locking pins, fasteners, slides, and mountings on plug-in assemblies.

(f) Size and form of special threads.

4.15 **Acceptance inspection of preparation for delivery.** The minimum inspection performed by the contractor shall be as specified in 4.15.1 and 4.15.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government — at its discretion — will verify, by product inspection, the contractor's compliance with specified requirements (see 6.6). Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.15.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table VI.

4.15.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be 10 percent. Classification of defects shall be as shown on table VII.

4.15.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed

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packs which are identical and will be submitted for acceptance at one time.

4.15.2.2 *Procedure in case of failure.* If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.15.2.3 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL–STD–105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

TABLE VI. *Preservation, and packaging and marking thereof*

Major

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

Minor

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Air not expelled from barrier.
4. Insufficient area of barrier material provided to permit 2 subsequent seals.
5. Any item of marking information other than 7 through 9 omitted, incorrect, or illegible.

TABLE VII. *Packing, and marking thereof*

Major

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

Minor

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL–STD–105.
4. Any item of required marking information other than 10 through 16 listed under major defects omitted, incorrect, or illegible.

4.16 Rough-handling test (preparation for delivery). When rough-handling test is required by the contract (see 6.2(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling (see Accuracy 4.8).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Bridge, Resistance ZM–4( )/U shall be preserved, packaged, and tested, in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described in the following paragraphs:

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5.1.1.1 *Technical manual.* Each technical manual shall be packaged method IC-3 as follows: Inclose each technical manual within a close-fitting box, type II, class b, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 *Unit package.*

5.1.1.2.1 *Bridge, Resistance ZM-4( )/U.* Each resistance bridge shall be unit packaged method IC-5 as follows: Close the unit and secure all fastenings, Cushion each unit on all surfaces with cells or pads or both, fabricated of double-faced corrugated fiberboard having a minimum dry bursting strength of 175 pounds designed to absorb the shock of impact encountered in handling and transit. The cells or pads shall be equal to the water-resistant requirements of the fiberboard box in which they are used. Place the cushioned unit within a close-fitting fiberboard box, style RSC, class 2 or 3 W5C conforming to the requirements of Specification PPP-B-636. Closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* Bridge, Resistance ZM-4( )/U shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration, and physical damage, during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* The bridges packaged as specified in 5.1 shall be packed in a close-fitting wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box or a fiberboard box conforming to Specification PPP-B-585, class 3, PPP-B-591 (overseas-type), PPP-B-601 style A or B, using plywood conforming to NN-P-515, type II or III, class 2, PPP-B-621 class 2 and PPP-B-636 class 2 or 3, respectively. Containers shall be closed and strapped with flat steel strapping in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 *Level B.* The bridges packaged as specified in 5.1 shall be packed in a close-fitting wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box or a fiberboard box conforming to Specification PPP-B-585 class 1, PPP-B-591 (domestic-type), PPP-B-601 (domestic-type), PPP-B-621 class 1 and PPP-B-636 class 1, respectively. Closure requirements shall be in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The bridges packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules, National Motor Freight Classification or Parcel Post Regulations, whichever is applicable

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Resistance Bridge ZM-4( )/U is intended for use in locating faults in communication circuits by Murray, Hilborn, or Varley loop resistance tests, or for locating opens by capacitance tests. Resistance may be measured from 1 milliohm to 1 megohm. Used as a resistance box, a range of from 0 to 10,110 ohms is obtained.

6.2 Order data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (see 3.5).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon — including rough-handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on figure 1 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required (see 3.7).

(i) Quantity of tools and running spare parts required (see 3.7).

(j) Facsimiles required (see MIL-T-945).

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: *ZM-4W/U. As soon as possible after* the award of the contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.3).

6.5 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of an accuracy, quality, and quantity that permits performance of the required acceptance inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

6.6 Verification inspection. The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A — The total of that inspection set forth in the quality assurance provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard MIL-STD-105.

(b) Type B — That inspection set forth in the quality assurance provisions of this specification or the contract

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reduced in amount under the reduced inspection provisions of Military Standard MIL-STD-105.

(c) Type C — A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the quality assurance provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Signal Corps
Navy—Bureau of Ships
Air Force—WADC

Preparing activity:

Army—Signal Corps

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PACKAGING, PACKING. AND MARKING PILOT PACK PROCESS SHEET

CONTRACTOR ORDER NO. _____

STREET ADDRESS EQUIPMENT _____

CITY AND STATE LEVEL _____

DATE PILOT PACK COMPLETED FOR EVALUATION _____

Item Name								
PACKAGING METHOD								
CLEANING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
PRESERVATION	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
CUSHIONING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
SECURING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
BARRIER	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
DESICCANT	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
UNIT CARTON	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
INTERMEDIATE CARTON	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
OUTER CARTON (CONSOLIDATED OR IIb)	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							
SECURE CLOSURE	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T	G		
	SIZE							

CASE LINER	SPECIFICATION	
	SEALING MATERIAL	
	SIZE	
SHIPPING CONTAINER	SPECIFICATION	
	STYLE	
	SIZE	
STRAPPING	SPECIFICATION	
	MATERIAL DESCRIPTION	
	SIZE	

LEGEND FOR MATERIAL: T—TYPE, G—GRADE C—CLASS

FIGURE 1

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